The History of HEMT

1979

The invention of the high-electron-mobility transistor (HEMT) is usually attributed to physicist Takashi Mimura, while working at Fujitsu in Japan. The basis for the HEMT was the GaAs (gallium arsenide) MOSFET which Mimura had been researching as an alternative to the standard silicon (Si) MOSFET since 1977. He conceived the HEMT in Spring 1979.[1] The HEMT was based on the concept of modulation doping first demonstrated by Ray Dingle and collaborators at Bell Labs in 1978. [2]



Takashi Mimura (M'73–SM'89–F'93) received the B.S. degree in physics from the Kwansei Gakuin University, Hyogo, Japan, in 1967, and the M.S. and Ph.D. degrees in solid-state physics from Osaka University, Osaka, Japan, in 1970 and 1982, respectively.

In 1970, he began his professional career as a Technical Staff Member with Fujitsu Ltd., Kobe, Japan, where his research centered around the development of microwave Si bipolar transistors and GaAs power MESFETs. In 1975 he joined Fujitsu Laboratories Ltd., Kawasaki, Japan, where he was involved with III–V compound semiconductor high-frequency high-speed devices including the heterostructure HEMT. He is currently a Fellow of Fujitsu Laboratories Ltd.

Dr. Mimura is a Fellow of the Institute of Electronics, Information and Communication Engineers (IEICE), Japan. He was the recipient of many awards for his pioneering work on HEMTs, including a 1998 Purple Ribbon Medal presented by the Japanese Government, the 1998 H. Welker Award presented at the International Symposium on Compound Semiconductors, the 1992 Imperial Invention Prize presented by the Japan Institute of Invention and Innovation, and the 1990 IEEE Morris N. Liebmann Memorial Award.

Modulation doping, a technique for fabricating semiconductors where free charge carriers are spatially separated from the donors, leading to high carrier mobilities, was first conceived at Bell Labs in 1977. The idea emerged from a conversation between Horst Störmer and Ray Dingle. Shortly afterward, Arthur Gossard implemented modulation doping.
"GaAs-AlxGa1-xAs superlattice structures in which electron mobilities exceed those of otherwise equivalent epitaxial GaAs as well as the Brooks-Herring predictions near room temperature and at very low temperatures are reported. This new behavior is achieved via a modulation-doping technique that spatially separates conduction electrons and their parent donor impurity atoms, thereby reducing the influence of ionized and neutral impurity scattering on the electron motion. We report the growth and properties of heterojunction superlattices of GaAs-$Al_xGa_{1-x}As$ in which the independent notion of modulation doping is incorporated. The carriers confined to the GaAs layer will form a pseudo-two dimensional electron gas (2DEG)" [2]

Their crude estimation has shown that high speed performance of HEMT should be 3 times than that of MESFET at 77K [1]

Mimura in his paper wrote about the first commercialization of his HEMT. "We, therefore, developed a prototype HEMT amplifier for satellite communications and introduced it at the 1983 International Solid-State Circuit Conference. When my colleague returned to his seat at the end of his talk, a conference attendee asked to buy the HEMT that had just been presented. That person was doing work for a radio astronomical observatory. The HEMT amplifier seemed efficient enough to detect weak microwave signals from dark nebula." [3]



Fig. 2. My colleague, Satoshi Hiyamizu, and Takashi Mimura testing the first HEMT IC.

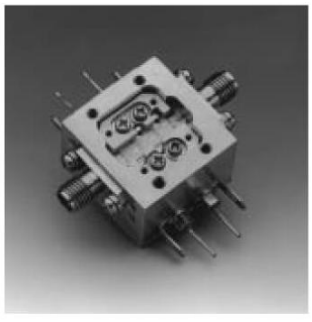


Fig. 3. First commercial HEMT: a cryogenic low-noise amplifier for the radio telescope at Nobeyama Radio Observatory, Nagano, Japan.

In the last 30 years, HEMTs have been demonstrated in several material systems, most notably AlGaAs/GaAs and AlGaN/GaN. Their uniqueness in terms of noise, power and high frequency operation has propelled HEMTs to gain insertion in a variety of systems where they provide critical performance value. 2DEG systems have also been a boon in solid-state physics where new and often bizarre phenomena have been discovered. [4]

1980

H.L Störmer and others first demonstrated P-type GaAs/AlGaAs MODFETS where 2D Hole Gas is used. They calculated the mobility to be 1700 cm/Vs at helium temperature.[5] In 1984, they published another paper mentioning that they have achieved a Transconductance of 28 mS/mm and 44 mS/mm, respectively for 77K and 4.2K in GaAs/(AlGa)As heterostructure interface.[6]
Since then, further studies has been done on P-type HEMTS involving other materials like InGaAs/AlGaAs, GaInAs/InP, etc. [7], [8], [9].

1993

One of the earliest mentions of a GaN-based HEMT is in the 1993 Applied Physics Letters article, by Asif Khan achieving transconductance of 28 mS/mm at 300 K and 46 mS/mm at 77 K. [10]

1999

The first time SiC was used in a HEMT.
"With the promise of a new level of power performance at microwave frequencies, GaN-based high-mobility-transistors (HEMTs) have attracted sustained research effort and shown steadfast improvements. Previous state-of-the-art included power densities of 5.3-6.9 W/mm at 10 GHz from small devices, as well as a total output power of 9.1 W at 7.4 GHz from a 3-mm-wide device. This performance was obtained using AlGaN/GaN HEMTs grown on SiC substrates for superior heat dissipation. However, these devices used a low Al mole fraction of 14% in the AlGaN layer, which had been shown to be less preferred in an earlier study with devices grown on sapphire substrates. In this work, high Al content AlGaN/GaN HEMTs on SiC substrates are demonstrated with remarkable performance enhancement." [11]

The History of Superconductivity

"When superconductivity was discovered by Onnes (1911),and for many years afterwards, it was thought to consist simply of a vanishing of all electrical resistance below the transition temperature. A major advance was the discovery of the Meissner effect (1933), which showed that a superconductor is a perfect diamagnet; magnetic flux is excluded from all but a thin penetration region near the surface. Not very long afterwards (1935), London and London proposed a phenomenological theory of the electromagnetic properties in which the diamagnetic aspects were assumed basic." [12]
Stating this in their paper in 1957, Bardeen, J., Cooper, L. N., & Schrieffer, J. R. proposed their Theory Of Superconductivity known as BCS theory.



Researches are conducted to combine the effect of superconductivity with the HEMT. The direct epitaxial growth of semiconductor heterostructures on top of crystalline superconductors has proved challenging.[13]

2018

The paper reports the successful use of molecular beam epitaxy to grow and integrate niobium nitride (NbN)-based superconductors with the wide-bandgap family of semiconductors—silicon carbide, gallium nitride (GaN) and aluminium gallium nitride (AlGaN). "We apply molecular beam epitaxy to grow an AlGaN/GaN quantum-well heterostructure directlyon top of an ultrathin crystalline NbN superconductor. The resulting high-mobility, two-dimensional electron gas in the semiconductor exhibits quantum oscillations, and thus enables a semiconductor transistor—an electronic gain element to be grown and fabricated directly on a crystalline superconductor. Using the epitaxial

superconductor as the sourceload of the transistor, we observe in the transistor output characteristics a negative differential resistance—a featureoften used in amplifiers and oscillators.
Our demonstration of the direct epitaxial growth of high-quality semiconductor heterostructures and devices on crystalline nitride superconductors opens up the possibility of combining the macroscopic quantum effects of superconductors with the electronic, photonic and piezoelectric properties of the group III/nitride semiconductor family.[13]

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